

2D:

• kartezični: $\vec{r} = x\hat{i} + y\hat{j}$

• polarni: $x = r\cos\varphi, y = r\sin\varphi$

• splošno: $x = f(x_\alpha, x_\beta) \rightarrow \vec{e}_{x_\alpha} = \frac{\partial \vec{r}}{\partial x_\alpha} \rightarrow \hat{e}_{x_\alpha} = \frac{\vec{e}_{x_\alpha}}{|\vec{e}_{x_\alpha}|}$
 $y = g(x_\alpha, x_\beta)$

$$d\vec{r} = \vec{e}_{x_\alpha} dx_\alpha + \vec{e}_{x_\beta} dx_\beta$$

• v splošnem $\vec{e}_{x_\alpha} \cdot \vec{e}_{x_\beta} \neq 0$! (nista $\perp$)

1) $\vec{e}_r, \vec{e}_\varphi, |\vec{e}_r|, |\vec{e}_\varphi|, \vec{e}_r \cdot \vec{e}_\varphi$

$$\vec{r} = (r\cos\varphi, r\sin\varphi)$$

$$\vec{e}_r = \frac{\partial \vec{r}}{\partial r} = (\cos\varphi, \sin\varphi) \quad |\vec{e}_r| = \sqrt{\cos^2\varphi + \sin^2\varphi} = 1$$

$$\vec{e}_\varphi = \frac{\partial \vec{r}}{\partial \varphi} = (-r\sin\varphi, r\cos\varphi) \quad |\vec{e}_\varphi| = r$$

$$\vec{e}_r \cdot \vec{e}_\varphi = -r\cos\varphi\sin\varphi + r\sin\varphi\cos\varphi = 0$$

$$T(r, \varphi) = \frac{1}{2} m \dot{\vec{r}}^2 = \frac{1}{2} m \dot{\vec{r}} \cdot \dot{\vec{r}} = \frac{1}{2} m (\dot{r}^2 + \dot{\varphi}^2 |\vec{e}_\varphi|^2) = \underline{\underline{\frac{1}{2} m (\dot{r}^2 + \dot{\varphi}^2 r^2)}}$$

$$\dot{\vec{r}} = \dot{r} \vec{e}_r + \dot{\varphi} \vec{e}_\varphi$$

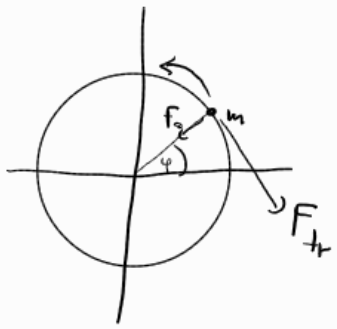
$$\ddot{\vec{r}} = \ddot{r} \vec{e}_r + \dot{r} \dot{\vec{e}}_r + \ddot{\varphi} \vec{e}_\varphi + \dot{\varphi} \dot{\vec{e}}_\varphi$$

$$\dot{\vec{e}}_r = \frac{d}{dt} (\cos\varphi, \sin\varphi) = (-\dot{\varphi} \sin\varphi, \dot{\varphi} \cos\varphi) = \frac{\dot{\varphi}}{r} \vec{e}_\varphi$$

$$\dot{\vec{e}}_\varphi = \dots = \frac{\dot{r}}{r} \vec{e}_\varphi - r \dot{\varphi} \vec{e}_r$$

$$\rightarrow \ddot{\vec{r}} = \ddot{r} \vec{e}_r + \dot{r} \frac{\dot{\varphi}}{r} \vec{e}_\varphi + \ddot{\varphi} \vec{e}_\varphi + \dot{\varphi} \left(\frac{\dot{r}}{r} \vec{e}_\varphi - r \dot{\varphi} \vec{e}_r \right) = \underline{\underline{\vec{e}_r (\ddot{r} - r \dot{\varphi}^2) + \vec{e}_\varphi (\ddot{\varphi} + \frac{2\dot{r}\dot{\varphi}}{r})}}$$

②



$r = R = \text{konst.}$

$\varphi(0) = \varphi_0$

$\dot{\varphi}(0) = v_{\varphi_0}$

$\vec{F}_{tr} = -\eta \vec{v} = -\eta \vec{r} \dot{\varphi} = -\eta \dot{\varphi} \vec{e}_\varphi$

$\vec{F}_o = |\vec{F}_o| \vec{e}_r$

$\vec{F} = m \vec{a}$

$\vec{F}_{tr} + \vec{F}_o = -\eta \vec{e}_\varphi \dot{\varphi} + |\vec{F}_o| \vec{e}_r = m \vec{r} = m \vec{e}_r r \dot{\varphi}^2 + m \vec{e}_\varphi \ddot{\varphi}$

$\rightarrow -\eta \dot{\varphi} = m \ddot{\varphi} \xrightarrow{z = \dot{\varphi}, \dot{z} = \ddot{\varphi}} -\eta z = m \dot{z}$

$|\vec{F}_o| = m r \dot{\varphi}^2$

$\frac{dz}{z} = -\frac{\eta}{m} dt$

$z = C e^{-\frac{\eta}{m} t}$

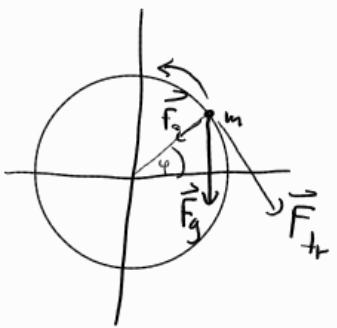
$z(0) = \dot{\varphi}(0) = v_{\varphi_0} \rightarrow C = v_{\varphi_0}$

$\rightarrow \dot{\varphi} = v_{\varphi_0} e^{-\frac{\eta}{m} t}$

$\varphi(t) - \varphi_0 = \int_0^t v_{\varphi_0} e^{-\frac{\eta}{m} t} dt = -\frac{m}{\eta} v_{\varphi_0} [e^{-\frac{\eta}{m} t} - 1]$

$\varphi(t) = \varphi_0 + \frac{m v_{\varphi_0}}{\eta} (1 - e^{-\frac{\eta}{m} t})$

③



$r = R = \text{konst.}$

$\varphi(-\frac{\pi}{2}) = \varphi_0$

$\delta \varphi(0) = \delta \varphi$

$\delta \dot{\varphi}(0) = \omega_0$

$\vec{F}_{tr} = -\eta \dot{\varphi} \vec{e}_\varphi$

$\vec{F}_o = |\vec{F}_o| \vec{e}_r$

$\vec{F}_g = -mg (\underbrace{\vec{e}_r \sin \varphi + \frac{\cos \varphi}{r} \vec{e}_\varphi}_{\hat{j}})$

$\vec{F}_{tr} + \vec{F}_o + \vec{F}_g = -\eta \dot{\varphi} \vec{e}_\varphi + |\vec{F}_o| \vec{e}_r - mg (\vec{e}_r \sin \varphi + \frac{\cos \varphi}{r} \vec{e}_\varphi) = m \vec{r} =$

$= m \vec{e}_r r \dot{\varphi}^2 + m \vec{e}_\varphi \ddot{\varphi}$

$\vec{e}_r: |\vec{F}_o| - mg \sin \varphi = m r \dot{\varphi}^2$

$\vec{e}_\varphi: -\eta \dot{\varphi} - mg \frac{\cos \varphi}{r} = m \ddot{\varphi} \rightarrow \text{stabilisiert gegen } -\frac{\pi}{2} \rightarrow \varphi = \varphi_0 + \delta \varphi$

$\cos \varphi \approx 0 - (-1) \delta \varphi = \delta \varphi$

$\rightarrow \delta \ddot{\varphi} + \underbrace{\frac{\eta}{m}}_{\frac{1}{2\beta}} \delta \dot{\varphi} + \underbrace{\frac{g}{r}}_{\omega_0^2} \delta \varphi = 0$

Nastavek: $\delta\varphi = A e^{\lambda t}$

$$\delta\dot{\varphi} = A\lambda e^{\lambda t} \rightarrow \cancel{A\lambda^2 e^{\lambda t}} + 2\beta\cancel{A\lambda e^{\lambda t}} + \omega_0^2\cancel{A e^{\lambda t}} = 0$$

$$\delta\ddot{\varphi} = A\lambda^2 e^{\lambda t}$$

$$\lambda_{1,2} = \frac{-2\beta \pm \sqrt{4\beta^2 - 4\omega_0^2}}{2} = -\beta \pm \sqrt{\beta^2 - \omega_0^2} = -\beta \pm i\Omega; \Omega^2 = \omega_0^2 - \beta^2$$

$$\rightarrow \delta\varphi = A e^{\lambda_1 t} + B e^{\lambda_2 t} = A \exp(-\beta + i\Omega)t + B \exp(-\beta - i\Omega)t$$

robní podmínky:

$$A + B = \delta\varphi_0 \quad A(-\beta + i\Omega) + B(-\beta - i\Omega) = \omega_0$$

$$(\delta\varphi_0 - B)(-\beta + i\Omega) + B(-\beta - i\Omega) = \delta\varphi_0(-\beta + i\Omega) - 2Bi\Omega = \omega_0$$

$$B = \frac{\omega_0 - \delta\varphi_0(-\beta + i\Omega)}{2i\Omega}, \quad A = \delta\varphi_0 - B = \frac{\delta\varphi_0(-\beta - i\Omega) - \omega_0}{2i\Omega}$$

...

Láze: $\delta\varphi = e^{-\beta t} (A e^{-i\Omega t} + B e^{i\Omega t}) = e^{-\beta t} (C \sin \Omega t + D \cos \Omega t)$

$$\rightarrow \delta\varphi(0) = D = \delta\varphi_0$$

$$\delta\dot{\varphi} = -\frac{1}{\rho} (C \sin \Omega t + D \cos \Omega t) e^{-\beta t} + (C \cos \Omega t - D \sin \Omega t) \Omega e^{-\beta t}$$

$$\rightarrow \omega_0 = C$$

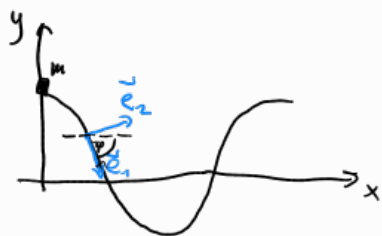
$$\rightarrow \varphi = \varphi_0 + e^{-\beta t} (\dots)$$

4. $y(x) = y_0 \cos \frac{x}{x_0}$

$y' = \frac{dy}{dx} \rightarrow \tan \varphi = y'$

$\hat{e}_1 = \hat{i} \cos \varphi + \hat{j} \sin \varphi$

$\hat{e}_2 = -\hat{i} \sin \varphi + \hat{j} \cos \varphi$



Ali se kladra odlepi?

$\vec{r} = x\hat{i} + y\hat{j}$

$\dot{\vec{r}} = \dot{x}\hat{i} + \dot{y}\hat{j} = v\hat{e}_1$

$\ddot{\vec{r}} = \dot{v}\hat{e}_1 + v\dot{\hat{e}}_1$

$y' = \frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{dy}{dx} \right) \frac{dx}{dt}$

$\dot{\hat{e}}_1 = -\hat{i} \sin \varphi \dot{\varphi} + \hat{j} \cos \varphi \dot{\varphi}$

$p = \arctan y' \rightarrow \dot{\varphi} = \frac{1}{1+(y')^2} y' = \frac{1}{1+(y')^2} \frac{d}{dx} \frac{dy}{dx} \frac{dx}{dt} = \frac{1}{1+(y')^2} y'' \dot{x}$

$v\hat{e}_1 = \dot{x}\hat{i} + \dot{y}\hat{j} = \dot{x}\hat{i} + y'\dot{x}\hat{j} \rightarrow \dot{x} = v \cos \varphi = \frac{v}{y'} \sin \varphi \rightarrow \dot{x} = v \cos \varphi = v \frac{1}{\sqrt{1+\tan^2 \varphi}} =$

$\dot{y} = \frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}$

$y' = -\frac{y}{x_0} \sin \frac{x}{x_0}, y = y_0 \cos \frac{x}{x_0}$

$\hat{j} = \hat{e}_1 \sin \varphi + \hat{e}_2 \cos \varphi$

$= \frac{v}{\sqrt{1+y'^2}}$

$\dot{x} = \frac{v}{y'} \sin \varphi =$

$= \frac{v}{y'} \frac{1}{\sqrt{1+\frac{1}{\tan^2 \varphi}}} = \frac{v}{y'} \frac{1}{\sqrt{1+y'^2}}$

$\rightarrow m\ddot{\vec{r}} = -\hat{j}mg + \hat{e}_2 |F_p|$

$m(\dot{v}\hat{e}_1 + v\dot{\hat{e}}_1) = m(\dot{v}\hat{e}_1 + v\dot{\hat{e}}_2 \frac{1}{1+y'^2} y'' \dot{x}) = -(\hat{e}_1 \sin \varphi + \hat{e}_2 \cos \varphi)mg + \hat{e}_2 |F_p|$

$\hat{e}_1:$

$m\dot{v} = -\sin \varphi mg$

$\hat{e}_2:$

$mv \frac{1}{1+y'^2} y'' \dot{x} = |F_p| - mg \cos \varphi$

Kladra se odlepi, ko $|F_p| = 0$

$\Rightarrow v^2 \frac{1}{1+y'^2} y'' \dot{x} = -g \cos \varphi \quad (\dot{x} = v \cos \varphi)$

$v^2 \frac{1}{1+y'^2} y'' = -g$

$W_k + W_p = W_{p0}$

$\frac{mv^2}{2} + mgy = mgy_0$

$v^2 y'' = -g y'^2 - g \leftarrow v^2 = 2g(y_0 - y)$

$2(y_0 - y)y'' + y'^2 + 1 = 0$

$\rightarrow 2(y_0 - y_0 \cos \frac{x}{x_0}) \left(-\frac{y_0}{x_0^2} \cos \frac{x}{x_0} \right) + \left(\frac{y_0}{x_0} \sin^2 \frac{x}{x_0} \right) + 1 = 0$

$2(1 - \cos \frac{x}{x_0}) \left(-\cos \frac{x}{x_0} \right) + \sin^2 \frac{x}{x_0} + 1 = 0$

$2\cos^2 \frac{x}{x_0} - 2\cos \frac{x}{x_0} + \sin^2 \frac{x}{x_0} + 1 = 0$

$\cos^2 \frac{x}{x_0} - 2\cos \frac{x}{x_0} + 2 = 0$

$\cos \frac{x}{x_0} = a \rightarrow a^2 - 2a + 2 = 0$ Nima R rešitev $\rightarrow$ kladra se ne odlepi

VRTECI SISTEMI

$$\vec{r}' = \vec{v}_{rel} + \vec{\Omega} \times \vec{r}$$

$$\ddot{\vec{r}}' = \ddot{\vec{a}}_{rel} + 2\vec{\Omega} \times \vec{v}_{rel} + \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$



$$\vec{r}'(0) = x_0 \hat{i} + y_0 \hat{j}$$

$$\dot{\vec{r}}'(0) = 0$$

$$\vec{r}'(t) = ?$$

$$m\ddot{\vec{r}} = \vec{F}_c + \vec{F}_{cor} = m\ddot{\vec{a}}_c + m\ddot{\vec{a}}_{cor} = -m(\vec{\omega} \times (\vec{\omega} \times \vec{r}) + 2\vec{\omega} \times \dot{\vec{r}})$$

$$\vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \omega \\ x & y & 0 \end{vmatrix} = -\omega y \hat{i} + \omega x \hat{j}$$

$$\vec{\omega} \times (\vec{\omega} \times \vec{r}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \omega \\ -\omega y & \omega x & 0 \end{vmatrix} = -\omega^2 x \hat{i} - \omega^2 y \hat{j} = -\omega^2 \vec{r}$$

$$\ddot{x} \hat{i} + \ddot{y} \hat{j} = \omega^2 \vec{r} + 2\omega \dot{y} \hat{i} - 2\omega \dot{x} \hat{j}$$

$$\left. \begin{array}{l} \hat{i}: \ddot{x} = \omega^2 x + 2\omega \dot{y} \\ \hat{j}: \ddot{y} = \omega^2 y - 2\omega \dot{x} \end{array} \right\} z = x + yi$$

$$\hat{i} + i\hat{j}: \ddot{z} = \omega^2 z - 2\omega(i\dot{x} - \dot{y}) = \omega^2 z - 2\omega \dot{z}i \rightarrow z = Ae^{\lambda t}$$

$$A\lambda^2 e^{\lambda t} = \omega^2 A e^{\lambda t} - 2\omega A \lambda e^{\lambda t} i$$

$$\lambda^2 + 2\omega i \lambda - \omega^2 = 0$$

$$\lambda_{1,2} = \frac{-2\omega i \pm \sqrt{4\omega^2 - 2\omega^2}}{2} = -i\omega \rightarrow z = (A + Bt)e^{-i\omega t}$$

$$\vec{r}'(t=0) = (x_0, y_0) \rightarrow x_0 + iy_0$$

$$\dot{\vec{r}}'(t=0) = 0 \rightarrow 0$$

$$z(t=0) = A = x_0 + iy_0$$

$$\dot{z} = -A i \omega e^{-i\omega t} + B e^{-i\omega t} - B t i \omega e^{-i\omega t}$$

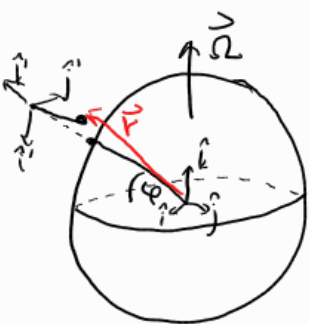
$$\dot{z}(t=0) = -i\omega A + B = 0$$

$$B = i\omega(x_0 + iy_0) = -\omega y_0 + i\omega x_0$$

$$\begin{aligned} \rightarrow z &= (x_0 + iy_0 - \omega y_0 t + i\omega x_0 t) e^{-i\omega t} = ((x_0 - y_0 \omega t) + i(x_0 \omega t + y_0)) (\cos \omega t - i \sin \omega t) = \\ &= (x_0 - y_0 \omega t) \cos \omega t + i(x_0 \omega t + y_0) \cos \omega t - i(x_0 - y_0 \omega t) \sin \omega t + (x_0 \omega t + y_0) \sin \omega t \end{aligned}$$

$$\left. \begin{aligned} x' &= \operatorname{Re}(z) = (x_0 - y_0 \omega t) \cos \omega t + (x_0 \omega t + y_0) \sin \omega t \\ y' &= \operatorname{Im}(z) = (x_0 \omega t + y_0) \cos \omega t - (x_0 - y_0 \omega t) \sin \omega t \end{aligned} \right\} \text{ spirala}$$

Focaultovo nihalo



$$m \vec{a}_{\text{rel}} = \sum \vec{F} - 2m \vec{\Omega} \times \vec{v}_{\text{rel}} - m \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

$$\vec{r} = \vec{R} + \vec{r}'$$

$$\sum \vec{F} = \vec{F}_g + \vec{F}_v$$

približek: $\vec{F}_g = -mg \hat{k}$ (razlikuje se za nek majhen kot)

$$\vec{F}_v = -F_v \frac{\vec{r}'}{|\vec{r}'|} \approx -\frac{F_v}{l} \vec{r}'$$



-Ravnovesna lega glede na S' : $\vec{a}_{\text{rel}} = 0$, $\vec{v}_{\text{rel}} = 0$, $\vec{r}_0'$

$$\rightarrow 0 = -mg \hat{k} - \frac{F_v}{l} \vec{r}_0' - m \vec{\Omega} \times (\vec{\Omega} \times \vec{R}) - m \vec{\Omega} \times (\vec{\Omega} \times \vec{r}_0')$$

zabeharimo

$$\vec{r}_0' = \frac{l}{F_v} [-mg \hat{k} - m \vec{\Omega} \times (\vec{\Omega} \times \vec{R})]; \quad \vec{r}' = \vec{r}_0' + \vec{\alpha} \quad \text{majhni odmiki}$$

$$\rightarrow m \vec{a}_{\text{rel}} = -mg \hat{k} - \frac{F_v}{l} (\vec{r}_0' + \vec{\alpha}) - 2m \vec{\Omega} \times \vec{v}_{\text{rel}} - m \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

$$m \vec{a}_{\text{rel}} = -\frac{F_v}{l} \vec{\alpha} - 2m \vec{\Omega} \times \vec{v}_{\text{rel}}, \quad \vec{\alpha} = x \hat{i}' + y \hat{j}' + z \hat{k}', \quad \text{približek: } z=0, F_v \approx mg$$

$$\ddot{x} \hat{i}' + \ddot{y} \hat{j}' = -\frac{g}{l} [x \hat{i}' + y \hat{j}'] + 2(\dot{y} \Omega \sin \varphi \hat{i}' - \dot{x} \Omega \sin \varphi \hat{j}') \quad \vec{v}_{\text{rel}} = \dot{x} \hat{i}' + \dot{y} \hat{j}'$$

$$\hat{i}': \ddot{x} = -\frac{g}{l} x + 2\Omega \dot{y} \sin \varphi$$

$$\hat{j}': \ddot{y} = -\frac{g}{l} y - 2\Omega \dot{x} \sin \varphi$$

$$\vec{\Omega} \times \vec{v}_{\text{rel}} = \begin{vmatrix} \hat{i}' & \hat{j}' & \hat{k}' \\ 0 & \Omega \cos \varphi & \Omega \sin \varphi \\ \dot{x} & \dot{y} & 0 \end{vmatrix} =$$

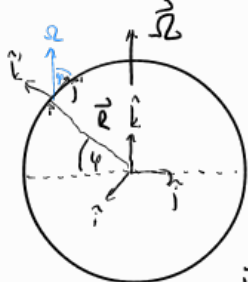
$$-\dot{y} \Omega \sin \varphi \hat{i}' - \Omega \cos \varphi \dot{x} \hat{k}' + \dot{x} \Omega \sin \varphi \hat{j}'$$

$$\ddot{z} = -\frac{g}{l} z + 2\Omega \sin \varphi (\dot{y} - i \dot{x}) = -\omega_0^2 z - i 2\Omega \sin \varphi \dot{z}$$

$$\downarrow$$

$$g = \omega_0^2 l$$

$$(y - ix)i = x + iy$$



$$\vec{r}'(0) = 0$$

$$\dot{\vec{r}}'(0) = 0$$

$$\vec{r}'(t) = ?$$

$$m\ddot{\vec{r}}' = -mg\hat{k}' - 2m\vec{\Omega} \times \vec{v}' - m\vec{\Omega} \times (\vec{\Omega} \times (\vec{r} + \vec{r}'))$$

$$\ddot{x}\hat{i}' + \ddot{y}\hat{j}' + \ddot{z}\hat{k}' = -g\hat{k}' - 2\Omega(\cos\varphi\dot{z}\hat{i}' - \sin\varphi\dot{y}\hat{i}'' + \sin\varphi\dot{x}\hat{j}' - \cos\varphi\dot{x}\hat{k}')$$

$$\vec{r}' = x\hat{i}' + y\hat{j}' + z\hat{k}'$$

$$\vec{\Omega} \times \vec{v}' = \begin{vmatrix} \hat{i}' & \hat{j}' & \hat{k}' \\ 0 & \Omega\cos\varphi & \Omega\sin\varphi \\ \dot{x} & \dot{y} & \dot{z} \end{vmatrix} = \begin{bmatrix} \Omega\cos\varphi\dot{z} - \Omega\sin\varphi\dot{y} \\ \Omega\sin\varphi\dot{x} \\ -\Omega\cos\varphi\dot{x} \end{bmatrix}$$

$$\hat{i}': \ddot{x} = -2\Omega(\cos\varphi\dot{z} - \sin\varphi\dot{y})$$

$$\hat{j}': \ddot{y} = -2\Omega\sin\varphi\dot{x}$$

$$\hat{k}': \ddot{z} = 2\Omega\cos\varphi\dot{x} - g$$

$$z'' = \cos\varphi\dot{z} - \sin\varphi\dot{y}$$

$$y'' = \sin\varphi\dot{z} + \cos\varphi\dot{y}$$

$$\rightarrow \ddot{x} = -2\Omega\dot{z}$$

$$\ddot{z}'' = 2\Omega(\cos^2\varphi\dot{x} + \sin^2\varphi\dot{x}) - g\cos\varphi = 2\Omega\dot{x} - g\cos\varphi$$

$$\ddot{y}'' = 2\Omega(\dot{x}\sin\varphi\cos\varphi - \dot{x}\sin\varphi\cos\varphi) - g\sin\varphi = -g\sin\varphi$$

$$\ddot{y}'' = -g\sin\varphi \rightarrow \dot{y}'' = -g\sin\varphi t + C_1 \rightarrow y'' = -\frac{gt^2}{2}\sin\varphi + C_1t + C_2$$

$$\begin{aligned} y''(0) &= 0 \\ \dot{y}''(0) &= 0 \end{aligned} \Rightarrow C_1, C_2 = 0$$

$$\alpha = x + iz''$$

$$i\alpha = ix - z''$$

$$\Rightarrow \ddot{\alpha} = -ig\cos\varphi + 2i\Omega\dot{\alpha}$$

$$\ddot{\alpha} - 2i\Omega\dot{\alpha} = -ig\cos\varphi$$

$$1) \alpha_h: \ddot{\alpha}_h - 2i\Omega\dot{\alpha}_h = 0$$

$$\alpha_h = C_1 e^{2\Omega i t} + C_2$$

$$\lambda^2 - 2i\Omega\lambda = 0$$

$$\lambda_1 = 0 \quad \lambda_2 = 2\Omega i$$

$$2) \alpha_p$$

$$\text{Lagrange's}$$

$$\alpha_p = At + B \rightarrow -2i\Omega A = -ig\cos\varphi$$

$$A = \frac{g\cos\varphi}{2\Omega}$$

$$\rightarrow \alpha = C_1 e^{2\Omega i t} + \frac{g\cos\varphi}{2\Omega} t + C_2$$

$$\alpha(0) = 0 \rightarrow C_1 + C_2 = 0$$

$$\dot{\alpha}(0) = 0 \rightarrow 2i\Omega C_1 + \frac{g\cos\varphi}{2\Omega} = 0 \rightarrow C_1 = i \frac{g\cos\varphi}{4\Omega^2} \rightarrow C_2 = -i \frac{g\cos\varphi}{4\Omega^2}$$

$$\alpha = \frac{g\cos\varphi}{2\Omega} \left(\frac{i}{2\Omega} e^{2i\Omega t} + t - \frac{i}{2\Omega} \right) = i \frac{g\cos\varphi}{4\Omega^2} (\cos(2\Omega t) + i\sin(2\Omega t)) + \frac{g\cos\varphi}{2\Omega} t - \frac{i g\cos\varphi}{4\Omega^2}$$

$$x = \text{Re}(\alpha) = \frac{g\cos\varphi}{2\Omega} \left(t - \frac{\sin(2\Omega t)}{2\Omega} \right)$$

$$z'' = z\cos\varphi - y\sin\varphi \quad z''\cos\varphi + y''\sin\varphi = z$$

$$y'' = z\sin\varphi + y\cos\varphi \quad -z''\sin\varphi + y''\cos\varphi = y$$

$$\rightarrow z = \frac{g\cos\varphi}{4\Omega^2} (\cos(2\Omega t) - 1) - \frac{gt^2}{2} \sin 2\varphi$$

$$y = \frac{g\cos\varphi \sin\varphi}{4\Omega^2} (1 - \cos(2\Omega t)) - \frac{gt^2}{2} \sin\varphi \cos\varphi$$

Prost delec v polarnih koordinatah

$$\vec{a} = (\ddot{r} - \dot{\varphi}^2 r) \hat{e}_r + (\dot{\varphi} r + 2\dot{r}\dot{\varphi}) \hat{e}_\varphi = 0$$

$$r(t), \varphi(t) = ?$$

$$1) \ddot{r} = \dot{\varphi}^2 r$$

$$2) \ddot{\varphi} r + 2\dot{r}\dot{\varphi} = 0$$

$$(r^2 \dot{\varphi})' = 2r\dot{r}\dot{\varphi} + r^2 \ddot{\varphi} = r \cdot 2)$$

$$\downarrow$$

$$\lambda = r^2 \dot{\varphi}$$

$$1) \ddot{r} = \left(\frac{\lambda}{r^2}\right)^2 r = \frac{\lambda^2}{r^3}$$

$$\downarrow$$

$$\frac{d}{dt}(\dot{r}^2 + \frac{\lambda^2}{r^2}) = 2\dot{r}\ddot{r} - 2\frac{\lambda^2}{r^3}\dot{r} = \frac{\dot{r}}{2} \cdot 1 \rightarrow \Sigma = \dot{r}^2 + \frac{\lambda^2}{r^2}$$

$$\begin{cases} u = \Sigma r^2 - \lambda^2 \\ du = 2\Sigma r dr \end{cases}$$

$$\left(\frac{dr}{dt}\right)^2 = \Sigma - \frac{\lambda^2}{r^2} \rightarrow \int dt = \int \frac{dr \cdot r}{\sqrt{\Sigma - \frac{\lambda^2}{r^2}}} = \int \frac{r dr}{\sqrt{\Sigma r^2 - \lambda^2}} = \frac{1}{2\Sigma} \int \frac{du}{\sqrt{u}} = \frac{\sqrt{u}}{\Sigma} + C_1$$

$$\rightarrow t - t_0 = \frac{\sqrt{\Sigma r^2 - \lambda^2}}{\Sigma} \rightarrow r = \sqrt{(t - t_0)^2 \Sigma + \frac{\lambda^2}{\Sigma}}$$

$$\lambda = r^2 \dot{\varphi} \rightarrow \dot{\varphi} = \frac{\lambda \Sigma}{(t - t_0)^2 \Sigma + \lambda^2} = \frac{d\varphi}{dt} \rightarrow \varphi = \int \frac{\lambda \Sigma}{(t - t_0)^2 \Sigma + \lambda^2} dt = \frac{\lambda}{\Sigma} \int \frac{dt}{(t - t_0)^2 + \left(\frac{\lambda}{\Sigma}\right)^2} =$$

$$= \arctg \frac{\Sigma(t - t_0)}{\lambda} + C_2$$

$$\varphi - \varphi_0 = \arctg\left(\frac{\Sigma(t - t_0)}{\lambda}\right)$$

Kakšno gibanje je? ($\varphi_0 = 0$)

$$\begin{cases} 1 + \tan^2 x = \frac{1}{\cos^2 x} \rightarrow \cos x = \frac{1}{\sqrt{1 + \tan^2 x}} \\ \rightarrow \cos(\arctg x) = \frac{1}{\sqrt{1 + x^2}}, \sin(\arctg x) = \frac{x}{\sqrt{1 + x^2}} \end{cases}$$

$$\vec{r} = r \cos \varphi \hat{i} + r \sin \varphi \hat{j} = \sqrt{(t - t_0)^2 \Sigma + \frac{\lambda^2}{\Sigma}} \left(\cos\left(\arctg \frac{\Sigma(t - t_0)}{\lambda}\right) \hat{i} + \sin\left(\arctg \frac{\Sigma(t - t_0)}{\lambda}\right) \hat{j} \right) =$$

$$= \sqrt{\frac{(t - t_0)^2 \Sigma + \frac{\lambda^2}{\Sigma}}{1 + \left(\frac{\Sigma(t - t_0)}{\lambda}\right)^2}} \left(\hat{i} + \frac{\Sigma(t - t_0)}{\lambda} \hat{j} \right) = \sqrt{\frac{((t - t_0)^2 \Sigma + \lambda^2) \lambda^2}{\Sigma(\lambda^2 + \Sigma^2(t - t_0)^2)}}$$